

news feed

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hypothesis

USB OTG

■ What is it?

A supplement to the already existing USB 2.0 specification. With this standard, USB devices will be able to directly connect to other USB devices. Being backward compatible with prior USB standards, devices based on USB OTG will be able to operate on all PCs equipped with a USB interface.

■ How does it work?

When connected to a computer, a USB OTG device behaves as any conventional USB 2.0 device would. When two USB OTG devices are linked together, the cable sets a default host and a peripheral device. However, if the nature of the application decides that their roles need to be reversed, then a protocol called the Host Negotiation Protocol (HNP) changes the operating status of the connected devices.

■ What does this mean to me?

A digicam could interface directly with a photo printer or another camera to exchange pictures. A scanner could join forces with a network printer for sending faxes. These combinations are not possible today because USB has a PC-centric tree topology. The PC is the host (tree) and the USB devices are the branches. USB OTG will allow devices to communicate directly without a PC.

Anonymous P2P

Few of us are unaware of the legendary success of peer-to-peer (P2P) services like Napster and Audionet, that enabled millions worldwide to trade movies, songs and data freely. A group of veteran hackers revealed a new file sharing standard that promises to take P2P to the next level by increasing security. Developed by a 23-year-old hacker called the Mixer, this protocol is called Six/Four (named after the June 4, 1989 massacre at Tiananmen Square). Six/Four is based upon P2P

technologies coupled with VPN (Virtual Private Networks) and 'open proxy' networking in order to mask online identities. Traditional VPNs, used even with operating sys-

the network before finally reaching the destination on the Internet. The greater the number of servers between a user and the destination, the harder it is to trace the user's IP address. Six/Four implements about 100 steps of

layered anonymity before it reaches the final destination making it virtually impossible to trace the user. This new protocol was unveiled at the H2K2 hacker conference held between July 12 and 14 in New York City.

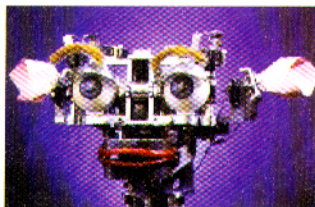


ILLUSTRATIONS: Mahesh Benkar

tems like Windows, establish a secure tunnel between two computers where information can be securely transferred across the Net. In Six/Four, the tunnel follows many computers in

Mushy robots

Cynthia Breazeal, an MIT professor, has built the 'sociable' robot called Kismet, with a cute face capable of smiling, frowning, looking surprised, and mimicking a few other basic emotions.



Kismet responds to the tone of voice, facial expressions and physical behaviour and comes up with

an appropriate response. Kismet is much like a child, and is being developed using theories of developmental psychology. The goal is to create a self-learning robot that can pick up emotions just like a growing child and learn from its surroundings.

Traditionally, robots have been designed to operate in hazardous and hostile environments, but now the applications are shifting towards entertainment and health care, where robots have to interact with people and play a part in their daily lives. Kismet promises to be a big step in this direction.

Palm OS 5 unveiled

PalmSource Inc has begun shipping the final version of Palm OS 5 to licensees and developers who can now adopt the highest-performance ARM chips from Intel, Motorola and Texas Instruments.

Palm OS 5 will deliver an enhanced user interface that enables increased personalisation, security, multimedia and wireless networking, while maintaining compatibility with today's Palm OS-based software and hardware.

